Week 11: Likelihood and Maximum Likelihood Estimation in More Complex Situations

F79MA Statistical Models A

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Introduction

In your studies of likelihood-based estimation, most of the examples you will have seen will take the form of calculating and maximising likelihoods based on random samples $x_1, .., x_n$ drawn from a statistical model with parameter θ and probability mass function $f_X(x; \theta)$. Often there will be an analytical solution for the maximum-likelihood estimator that you can derive using standard calculus techniques. Unfortunately, real life very often isn't that simple. In practical sampling situations (where you collect data that are believed to be generated from a model), there will often be additional sources of complexity that make maximum likelihood estimation non-trivial. Some of the sources of complexity include the following:

- There may be additional sources of variation in the measurements on top of the variation that arises from the statistical model.
- There may be 'censoring' of measurements (i.e. the exact value of measurements falling above or below a given threshold is not known).
- There may be dependence between measurements - in contrast to the usual independence assumptions.

When faced with these sources of complexity, we can still use likelihood as a method of estimating parameters. However:

- Deriving likelihoods won't necessarily be a straightforward process of writing down products of probability functions!
- You may have to work with additional parameters in your likelihood beyond those that you're actually interested in (these are known as nuisance parameters).
- You may end up with likelihoods that cannot be maximised analytically. You may need to resort to numerical methods of maximisation.

In the following sections, we will look at a number of situations where some of the above difficulties arise. You will get the chance to use 'R' to implement some basic techniques of optimisation.

Some examples

Measurement noise in an instrument

An engineer uses an electronic set of scales to investigate the distribution of the mass of silica beads used to pack a chromatography column. It is believed that the mass of the beads follow a $\mathrm{N}(\mu, \sigma^2)$ distribution, where μ and σ^2 are unknown. She selects a random sample of n beads and measures their masses using the device as $y_1, \ldots, y_n$. However, it is known that the measuring device introduces an error in each measurement of size e_i, where the errors are independent over measurements, are independent of the mass of the beads, and are distributed according to $\mathrm{N}(0, \gamma^2)$, where γ^2 is known. What are the maximum likelihood estimates of μ and σ^2?

Solution: To solve this problem, we simply note that the observations $Y_i \sim \mathrm{N}(\mu, \sigma^2+\gamma^2)$. Now if we transform the variance parameter and write $\alpha^2 = \sigma^2 + \gamma^2$, then we can immediately see that that the values $\hat{\mu}$ and $\hat{\alpha}^2$ that would maximise our usual normal likelihood are

$$\hat{\mu} = \bar{y}$$

and

$$\hat{\alpha}^2 = \frac{1}{n}\sum_{i=1}^{n}(y_i - \bar{y})^2.$$

From this we may deduce that the likelihood is maximised by setting $\hat{\sigma}^2 = \hat{\alpha}^2 - \gamma^2$ - or is it? A key thing to note here is that the range of the parameter α^2 is actually (γ^2, ∞), as opposed to the usual range of $(0, \infty)$. This means that an estimate of less than γ^2 does not make any sense. In the case of a data set for which $\hat{\alpha}^2 < \gamma^2$, then the maximum likelihood estimate of σ^2 would be 0. This can be seen by considering a graph of the log-likelihood as a function of σ^2 for $\mu = \bar{y}$.

It is very important to take into account the range of a transformed parameter when considering maximum likelihood estimation, and to be alive to the possibility that the estimate derived from unconstrained maximisation of the likelihood may, in fact, lie outside the permissible range. Failure to take this into account would lead one to reporting negative estimates of the variance σ^2 when $\hat{\alpha}^2 < \gamma^2$! (This would be seen as a major error in an exam, for example!!)

Non-analytic solutions

In many real situations, you will find that MLE's cannot be derived analytically and numerical techniques will be required in order to find the solution. This often arises in the context of non-standard situations with censoring and other such features.

Consider a population of containers each of which contains a number of identical weights. The number of weights in a container N is distributed as a Poi(λ) distribution. If a container contains 0 or 1 weight, it will float on water; otherwise it will sink. Out of a random sample of n containers, we observed that r of them float. What is the maximum likelihood estimate of λ?

Solution: Denote by p the probability that a container floats. Note that p is calculated by

$$p = \mathbf{P}(N \leq 1) = e^{-\lambda}(1 + \lambda).$$

Noting that p is a 1-1 function of λ (prove this!), we can immediately write down that the MLE of λ satisfies

$$\hat{p} = e^{-\hat{\lambda}}(1 + \hat{\lambda}).$$

It is worth recognising that the observation r is a realisation of a Bin(n, p) experiment. Therefore using results from the lectures we deduce that

$$\hat{p} = \frac{r}{n}.$$

The problem is then to solve

$$f(\lambda) = e^{-\lambda}(1 + \lambda) - \frac{r}{n} = 0.$$

This cannot be done analytically! We must use numerical techniques.

Finding roots using Newton-Raphson iteration: This is a straightforward iterative procedure which can be easily applied in many cases to find roots of equations of the form $f(x) = 0$. The main steps are to find an initial guess for the solution, and then to update this guess by approximating the graph of $f(x)$ as a straight line in a neighbourhood of the root.

If our initial guess is x_0, then it can be easily seen that the next update, x_1, is

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}.$$

Repeating this process, we can generate a sequence of updates $x_1, x_2, x_3, ...$, where the stopping criterion is when $f(x_i)$ becomes tolerably small.

Suppose now that in the above example, we have $n = 10$ and $r = 4$. Using Newton-Raphson iteration, we obtain the following recursion:

$$\lambda_{i+1} = \lambda_i + \frac{(e^{-\lambda_i}(1+\lambda_i) - 0.4)}{\lambda_i e^{-\lambda_i}},$$

where λ_0 is an initial guess.

We can implement this in 'R' to find the MLE estimate.

```
# R code to implement Newton-Raphson method
lam <- 2 # Select an initial guess for lambda


# Calculate the error
err <- function(lam) (exp(-lam)*(1+lam)-0.4)


# Now apply the update equation above repeatedly
# until the absolute value of the error is sufficiently small


while(abs(err(lam)) > 0.0000001) {
  lam = lam + err(lam)/(lam*exp(-lam))
}
```

Exercise: Implement this code in R and calculate the MLE of λ.

Asking sensitive questions

We consider a situation analogous to the above, where an underlying observed process is modified by a second random process arising in the case of binomial sampling. Suppose that we wish to investigate the proportion, p, of the student population who have cheated in an exam (hopefully negligibly small). We might take a random sample of size N, and ask them whether they have cheated. However, it is unlikely that they would all give truthful answers to this, since they may be suspicious that their responses are traced back to them. This situation arises in many surveys where answering a question may embarrass respondents. One way around this is to ask respondents to carry out the following task.

- First toss two fair coins and note the results.
- If TT the respondent answers 'yes', if HH the respondent answers 'no'.
- Otherwise the respondent gives a truthful answer to the question 'Have you ever cheated in an exam?'

The hope is that students answer truthfully, as a positive response would not necessarily indicate that the student had cheated.

Suppose that, out of the sample of N students, a total of r students respond 'yes' in the survey. What is the maximum likelihood estimate of p from these data?

Solution: Let q denote the probability that an individual gives a positive response. Then

$$q = \frac{1}{4} + \frac{p}{2},$$

or equivalently,

$$p = 2\left(q - \frac{1}{4}\right).$$

Note that the range of q is $[1/4, 3/4]$ corresponding to the range of $[0, 1]$ for p. Carrying out maximisation of the likelihood with respect to q, we may be tempted to write $\hat{q} = \frac{r}{N}$. However, this estimate is only valid if r lies in the range $[\frac{N}{4}, \frac{3N}{4}]$. If this is not the case then we must assign the maximum likelihood estimate to be the appropriate end point of the interval. Otherwise p would be estimated to be a nonsensical value.

Again a graph of the likelihood would demonstrate what is happening in situations where r falls outside the range.

Errors in reporting

In some situations we may be interested in estimating the rate of occurrence of a certain kind of event (crime, environmental disasters, insurance claims). However, the problem of estimation may be complicated by the fact that not all events are reported. For example, suppose that the number of industrial accidents in a certain factory follows a Poi(λ) distribution, where λ is unknown. Suppose further it is known that 20% of accidents are not reported and that failures to report occur independently of each other. In a given year, 10 accidents are reported. What is the MLE of λ?

Solution: To solve this problem, we will first show that the distribution of reported accidents follows a Poi($p\lambda$), where $p = 0.8$ is the probability that an accident is reported. This can be done in several ways, including from first principles. Let $Z \sim \text{Poi}(\lambda)$ denote the number of accidents and let X denote the number of accidents that are actually reported. The probability mass function of X is

$$
\begin{aligned}
\mathbf{P}(X = x) &= \sum_{y=x}^{\infty} \mathbf{P}(Z = y)\, \mathbf{P}(X = x \,|\, Z = y) \\
&= \sum_{y=x}^{\infty} \frac{e^{-\lambda}\lambda^y}{y!} \binom{y}{x} p^x (1-p)^{y-x} \\
&= \frac{e^{-\lambda}(\lambda p)^x}{x!} \sum_{y=x}^{\infty} \frac{\lambda^{y-x}}{(y-x)!} (1-p)^{y-x} \\
&= \frac{e^{-\lambda}(\lambda p)^x}{x!} \sum_{y=0}^{\infty} \frac{\lambda^y}{y!} (1-p)^y \\
&= \frac{e^{-\lambda}(\lambda p)^x}{x!} \sum_{y=0}^{\infty} \frac{(\lambda(1-p))^y}{y!} \\
&= \frac{e^{-\lambda}(\lambda p)^x}{x!} e^{\lambda(1-p)} \\
&= \frac{e^{-\lambda p}(\lambda p)^x}{x!}
\end{aligned}
$$

From this it follows from standard results on MLEs for the Poisson distribution that the MLE of $p\lambda$ is equal to 10. Therefore the MLE of λ is $\hat{\lambda} = \frac{10}{p} = 12.5$. Note that we are able to get an MLE for λ because p is known in this case. Suppose that p were not known and requires estimation. In this latter case, we could construct a likelihood function $L(p, \lambda)$ of the form

$$L(p, \lambda) \propto \exp(-\lambda p)(\lambda p)^{10}.$$

It is obvious that this likelihood is constant on any curve $\lambda p = c$, so that it is not possible to find a unique maximum likelihood estimate of (λ, p) over the parameter space.

Other forms of censoring

A very common situation, e.g. when fitting statistical models to failure times, survival times, *inter alia*, is to find that data measurement ceases before all the times have been observed. In these situations we can often derive a likelihood for the data and maximise it.

Consider a very simple situation. The lifetimes of a particular electronic component are believed to follow an Exp(λ) distribution. A random sample of n components are set in operation at time $t = 0$. Up until time $t = T$, the failure times of the components (assumed to be independent) are measured precisely. The experiment stops at time $t = T$, after which time k components have failed at times $0 < t_1 < t_2 < \ldots < t_k < T$, respectively, and the remaining $(n - k)$ components are still operational. How can we analyse these data using likelihood?

Solution: It is not appropriate to ignore the components that have not failed before time T and treat the k failures as a random sample from Exp(λ). *Why not?* We must construct a likelihood $L(\lambda)$ as the probability of the data given λ. We can derive this as follows:

$$L(\lambda) \propto \prod_{i=1}^{k} \lambda e^{-\lambda t_i} \prod_{i=k+1}^{n} e^{-\lambda T} = \lambda^k \exp\left(-\lambda \sum_{i=1}^{k} t_i\right) \left(e^{-\lambda T}\right)^{n-k},$$

i.e.

$$L(\lambda) \propto \lambda^k \exp\left\{-\lambda \left(\sum_{i=1}^{k} t_i + (n-k)T\right)\right\}.$$

This is easily maximised using standard techniques to give an estimate

$$\hat{\lambda} = \frac{k}{(n-k)T + \sum_{i=1}^{k} t_i}.$$

Does this look sensible? Note that this estimate can be interpreted as the total number of failures observed over the total amount of operating time in the experiment. Given the interpretation of λ as a failure rate, this is indeed a 'sensible-looking' estimate. Note that in the above analysis, the key point was that, for those components which did not fail by time T, the contribution to the likelihood was a factor of the form

$$\mathbf{P}(X > T; \lambda) = 1 - F_X(T; \lambda),$$

where F_X denotes the cumulative distribution function of the lifetime X.

Extension: Distribution of the MLE

In most situations that you will encounter (where the experiment involves the collection of a sufficient amount of data), the distribution of an MLE will be approximately normal with mean equal to the true parameter θ, and variance equal to the reciprocal of the information function, with analogous results for higher dimensional situations. At this point we will consider one-dimensional examples. In mathematical terms, this means that the MLE approximately satisfies:

$$\hat{\theta} \approx \mathrm{N}(\theta, I(\theta)^{-1})$$

The variance of the MLE under repeated sampling gives us a measure of *how informative the experiment is* about the parameter that we are trying to estimate. Intuitively, we should expect the variance to decrease as the amount of information from the experiment increases.

Continuing from the previous example: Suppose now that a second experimenter opts for a different design of experiment whereby she measures precisely all the failure times of the components. Intuitively, we should expect this second experiment (which provides all the information of the first and more) to give a more informative estimate of λ, as demonstrated by having a smaller variance. *Does this turn out to be the case?*

Let us first work out the asymptotic variance of the MLE for this second case where the data are uncensored. Suppose that $\tilde{\lambda}$ is the MLE of λ in this second case. We recall from exercises in class that $\mathbf{Var}(\tilde{\lambda}) = 1/I(\lambda) = \lambda^2/n$.

In the original case, where the observations are censored, it can be shown that the variance is increased by a factor of $(1 - e^{-\lambda T})^{-1}$, i.e., (*prove it!*)

$$\mathbf{Var}(\hat{\lambda}) = \frac{\lambda^2}{n(1 - e^{-\lambda T})}.$$

It is unsurprising that $\mathbf{Var}(\hat{\lambda}) \geq \lambda^2/n = \mathbf{Var}(\tilde{\lambda})$, since we would expect a larger amount of uncertainty to be associated with our parameter estimate in cases where data are censored. However, if we allow the experiment to go on forever (i.e., $T \to \infty$), we would have $\mathbf{Var}(\hat{\lambda}) \to \lambda^2/n$.

This makes intuitive sense. As T increases, the probability of any censoring decreases and the variance of the estimator approaches that for the completely uncensored case. Note that these results are asymptotic - they hold when the number of independent observations in the experiment (i.e. n) is sufficiently large.

Observations that are not identically distributed

Suppose we have a random sample $\underline{X} = (X_1, ..., X_n)$, where the sampling density of X_i is $f_i(x; \theta)$ (so that each observation is possibly drawn from a different density). If the observations $x_1, ..., x_n$ are independent, we can still form a likelihood function

$$L(\theta; \underline{x}) = \prod_i f_i(x_i; \theta)$$

and maximise it to estimate θ.

Consider the simple birth process where we have a population of individuals that grows in time according to the following rule: Each individual in the population gives birth to new individuals according a Poisson process with rate θ. That is, the times between births arising from any given individual are i.i.d. Exp(θ). Suppose now we observe a

population which starts with a single individual at time $t = 0$, and then subsequently observe the times at which the first n births occur to be $(t_1, t_2, ..., t_n)$.

We recode this set of observations as $x_1, x_2, ..., x_n$ such that

$x_1 = t_1,$

$x_2 = t_2 - t_1,$

...

$x_n = t_n - t_{n-1},$

so that $\underline{x} = (x_1, ..., x_n)$ represents the set of waiting times between the births, or from $t = 0$ to t_1 in the case of the first birth.

Exercise: Use the memoryless property of the exponential distribution to show that $X_i \sim \text{Exp}(i\theta)$ and that $X_1, ..., X_n$ are independent.

It is now straightforward to derive and maximise a likelihood function for this setting as

$$L(\theta; \underline{x}) \propto \theta^n \exp\left(-\theta \sum i x_i\right)$$

so that it can be easily shown that $\hat{\theta} = \frac{n}{\sum i x_i}$. Note that the same result can be derived even more quickly by noting that $Y_i = iX_i$ are i.i.d. $\text{Exp}(\theta)$ and then using the standard result for the MLE based on i.i.d. samples from the Exponential distribution.

A point for discussion. Suppose that in addition to the birth times, the data recorded the identity of each individual who had given birth. Would this provide any additional information on θ over and above that provided by observation of the birth times alone?